

Neurolabs Cordova SDK Integration Guide

Technical Integration Specification

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1. Scope

This guide is for partner hybrid apps integrating 'neurolabs-cordova-sdk'.

2. Changes

The Cordova plugin is a thin bridge over the native SDKs, so most changes listed here come from the underlying 'neurolabs-android-sdk' and 'neurolabs-ios-sdk' releases that the plugin pulls via 'scripts/prepare-sdk.js' at install time. No public JS API has been broken since v1.1.9.

v1.3.2

- Plugin pulls native SDK v1.3.2 binaries on 'cordova plugin add': Android AAR + iOS xcframework refreshed via the mobile-dist manifests.
- Native v1.3.2 highlights (no JS surface change required):
 - Mixpanel ingestion now lands in the EU region (events were previously POSTed to the US endpoint and silently dropped). iOS also fixes a separate bug where Mixpanel rejected every event because 'time' was serialized as a string rather than a JSON number.
 - Done button stays responsive while the SDK persists captures - disk I/O moved off the main thread on both platforms, inline spinner / progress indicator surfaced.
 - Android: native 'SIGSEGV' fix in 'com.google.ai.edge.litert.Model.nativeLoadAsset' when the TFLite asset is stored compressed in the APK; the loader probes the asset with 'AssetManager.openFd' first and extracts to internal storage when the asset is not mmap-able.
 - Onboarding barcode pipeline normalises to GTIN-14 end-to-end and propagates the scanned barcode format through every product-lookup client.
 - iOS: new 'NLOperationsCatalogClient' (in 'ProductAuditKit') for the operations catalog; replaces 'precondition'-based validation with thrown errors so a malformed barcode no longer crashes the host app.

v1.2.5

- Native sequence-counter parity: Cordova bridge surfaces the 'showsSequenceCounterLabel' (and 'showsCapturedCountLabel') options to match the native SDK toggles.
- Demo SDK setup builds fixed; iOS Sentry injection timing hardened; Android LiteRT dependency pinning correction; sequence-counter argument plumbed through correctly.

v1.2.3

- 'openCamera' supports 'cameraMode: "ar" | "default"' - ""default"" enables the 0.5/1 lens switcher on both iOS and Android.
- 'init' supports 'analyticsEnabled', 'sentryEnabled', 'sentryDsn' for controlling SDK analytics and crash reporting.

- Tightened default quality validation thresholds (blur, glare, perspective) across both platforms.

v1.2.2

- Internal: Android plugin integration handling refactor; no public JS API change.

v1.2.1

- Cordova install hooks hardened; demo integration updated.

v1.2.0

- Internal scaffolding for the v1.2.x line; no JS API change.

v1.1.9

- Init and per-session routing use 'taskUUID'.
- Native detector/model warmup is enforced in 'init' and guarded in 'openCamera'.
- 'openCamera' can return 'MODEL_INIT_FAILED' when native model initialization fails.
- 'autoCloseAfterCapture=true' now keeps preview enabled and closes after Save confirmation.
- Manual finish ('Done') is available when 'autoCloseAfterCapture=false' and 'allowManualFinish=true'.
- 'cameraClosed' event now includes 'message'.
- 'captureQueued' events are deferred while camera is open and flushed after 'cameraClosed'.

3. Install

Download the plugin package from

'<https://github.com/neurolaboratories/neurolabs-mobile-dist/releases>'. The install hooks download and wire up the native SDKs automatically - no manual AAR or xcframework placement needed.

Native baseline requirements:

- Android: 'minSdkVersion 26', 'compileSdkVersion 36', JDK 17, Android Gradle Plugin 8.9.1+ (the SDK's androidx dependencies require it). The plugin injects 'GradlePluginKotlinVersion 2.3.0' and raises the Java source/target compatibility to 17 on its own; override the Kotlin version only via a '<platform name="android">'-'scoped preference.
- iOS: deployment target iOS 17.0+, Xcode 16.4+ recommended (release toolchain baseline).

Add to 'config.xml' (cordova-android 14 defaults are lower):

```
<platform name="android">
  <preference name="android-minSdkVersion" value="26" />
  <preference name="android-compileSdkVersion" value="36" />
  <preference name="AndroidGradlePluginVersion" value="8.9.1" />
  <preference name="GradleVersion" value="8.13" />
</platform>
```

Android (Windows or macOS)

```
# Add the Android platform first, then install the plugin
cordova platform add android
cordova plugin add /path/to/neurolabs-cordova-sdk-vX.Y.Z.tgz
```

The 'before_plugin_install' hook downloads 'neurolabs-android-sdk.aar' from the matching GitHub release, stores it inside the plugin, and 'plugin.xml' copies it to 'app/libs/' automatically. 'neurolabs.gradle' wires it into the build via 'flatDir'.

iOS (macOS only)

```
# Add the iOS platform first, then install the plugin
cordova platform add ios
cordova plugin add /path/to/neurolabs-cordova-sdk-vX.Y.Z.tgz
```

The hook downloads 'NeurolabsSDK.xcframework' from the matching GitHub release and Cordova embeds it automatically during build. Requires 'curl' (standard on macOS) or 'gh' CLI. For private releases set 'GITHUB_TOKEN'.

The iOS install hook injects the 'Sentry' Swift Package dependency into the generated Xcode project automatically during platform preparation. If Xcode still reports 'unable to find module dependency: Sentry', re-run 'cordova prepare ios' so the package graph refreshes in the generated project.

Optional overrides (use a pre-downloaded file or a specific URL instead of auto-download):

```
# Local file
NEUROLABS_IOS_XCFRAMEWORK_ZIP=/path/to/NeurolabsSDK.xcframework-vX.Y.Z.zip \
  cordova plugin add /path/to/neurolabs-cordova-sdk-vX.Y.Z.tgz

NEUROLABS_ANDROID_AAR_PATH=/path/to/neurolabs-android-sdk-vX.Y.Z.aar \
  cordova plugin add /path/to/neurolabs-cordova-sdk-vX.Y.Z.tgz

# Direct URL
NEUROLABS_IOS_XCFRAMEWORK_URL=https://github.com/.../NeurolabsSDK.xcframework-vX.Y.Z.zip \
  cordova plugin add /path/to/neurolabs-cordova-sdk-vX.Y.Z.tgz

NEUROLABS_ANDROID_AAR_URL=https://github.com/.../neurolabs-android-sdk-vX.Y.Z.aar \
  cordova plugin add /path/to/neurolabs-cordova-sdk-vX.Y.Z.tgz
```

iOS + Android (macOS)

Add both platforms before installing the plugin - both hooks run in a single 'cordova plugin add'.

4. SDK Initialization + Warmup

```
const Neurolabs = cordova.require('ai.neurolabs.cordova.Neurolabs');

await Neurolabs.init({
  apiKey: '<API_KEY>',
  operationsApiKey: '<OPERATIONS_API_KEY>', // REQUIRED for uploads (see note below)
  // operationsBaseUrl: 'https://api.operations.staging.neurolabs.ai/v1', // optional override (staging / self-hosted)
  apiUrl: 'https://api.neurolabs.ai/v2',
  taskUUID: '<DEFAULT_TASK_UUID>',
  allowBase64PhotoExport: false,
  returnCaptureFileUri: false, // v1.6.5 opt-in file-URI streaming (see 8.1)
  analyticsEnabled: true, // SDK analytics (default true)
  sentryEnabled: true, // Sentry crash reporting (default true)
  // sentryDsn: 'https://...', // optional custom Sentry DSN
  // yoloModelVariant: 'shelf_rows', // see below
});
```

> **Upgrade note (pre-1.6 integrators): 'operationsApiKey' is now required for uploads.**

> > The legacy '/images' upload endpoint has been removed - the native > capture queue is operations-only. If you call 'init()' **without** > 'operationsApiKey', 'init()' and 'openCamera()' still succeed and photos > still queue, but **every upload fails silently**: the queue item errors > with 'OPERATIONS_KEY_REQUIRED' (non-retryable), surfaced only through the > 'uploadFailed' event. There is no console error and no init-time > rejection, so this is easy to miss during an upgrade. > > "js > Neurolabs.addListener('uploadFailed', ({ errorCode, message }) => { > if (errorCode === 'OPERATIONS_KEY_REQUIRED') { > console.error('Missing operationsApiKey in init() - uploads will never succeed.', message); > } > }); > " > > 'operationsBaseUrl' (optional) overrides the operations endpoint for > staging / self-hosted deployments. It must be an 'https' URL with a host > and no embedded credentials; 'http', malformed, or userinfo-bearing > values reject at 'init()' with 'INVALID_ARGUMENT'. Omit it to keep the > production default ('https://api.operations.neurolabs.ai/v1').

YOLO model variant

'yoloModelVariant' (optional, default 'nlb_model') picks the detector model the SDK loads. Accepted values, case-insensitive:

Value	Description
'nlb_model'	Legacy product detector (default).
'yolo_v26'	YOLOv2.6 product detector.
'yolo_v11_seg'	YOLO11-Seg product detector with mask coefficients.
'shelf_rows'	May-26 shelf-row detector - emits per-row regions + masks. Drives the row-aware live guidance (per-row quality chips, mask-based tilt/yaw, lock-on overlay). Requires the 'weights_may_26.tflite' / 'weights_may_26_labels.json' assets to be bundled.

Passing an unknown value raises 'INVALID_ARGUMENT' rather than silently falling back to 'nlb_model'.

Warmup is native-side; monitor progress through 'loadingProgress' event.

```
Neurolabs.addListener('loadingProgress', (payload) => {
  console.log('loading', payload);
});
```

5. Queue Management

```
await Neurolabs.setAutoSyncEnabled(true);
await Neurolabs.setWifiOnlyUploadsEnabled(false);
await Neurolabs.setDeletePhotosOnUploadSuccess(true);

await Neurolabs.pauseQueue();
await Neurolabs.resumeQueue();
await Neurolabs.setRetryPolicy({ retryCount: 5 });

const status = await Neurolabs.getQueueStatus();
console.log('queue status', status);

await Neurolabs.flushQueue();
await Neurolabs.retryFailedUploads();
```

6. Open Custom Camera

```
await Neurolabs.openCamera({
  sessionId: crypto.randomUUID(),
  type: 'shelf',
  cameraMode: 'default',           // 'ar' or 'default' - enables 0.5/1 lens switcher
  guidanceMode: 'strict',
  validationPreset: 'ios_parity',

  confidenceThreshold: 0.25,
  iouThreshold: 0.45,
  maxCaptures: 1,

  showAlignmentGuidance: true,
  enableValidation: true,
  liveQualityChecksEnabled: true,
  liveQualityTargetFps: 6,

  // keep custom-camera guidance UI path
  showDetections: false,
  showCapturedRegions: false,

  // optional metadata/cropping
  sendDetectionsMetadata: true,
  enablePreviewCropping: true,

  // now defaults to true if omitted, but explicit is clearer
  autoCloseAfterCapture: true,

  // manual-finish mode (Done button):
  // allowManualFinish: true,
  // minCapturesBeforeDone: 3,

  // per-session routing override
  taskUUID: 'bfc85982-b955-4f65-9f32-b6dbed85f364',

  // image size constraints
  maxImageDimension: 1920,
  maxImageWidth: 1920,
  maxImageHeight: 1920,
  imageCompressionQuality: 0.85
});
```

Manual-finish example:

```
await Neurolabs.openCamera({
  sessionId: crypto.randomUUID(),
  type: 'shelf',
  guidanceMode: 'strict',
  maxCaptures: 10,
  autoCloseAfterCapture: false,
  allowManualFinish: true,
  minCapturesBeforeDone: 3
});
```

6a. Multi-Bay Capture (wide shelves)

Multi-bay capture (native SDK v1.6.0+) guides the user across several overlapping "bays" of a wide shelf in a single session and de-duplicates detections across them. Enable it by passing a 'multiBay' options object to 'openCamera':

```
await Neurolabs.openCamera({
  sessionId: crypto.randomUUID(),
  type: 'shelf',
  guidanceMode: 'strict',
  multiBay: {
    minBays: 2, // 1..8
    maxBays: 4, // 2..8
    targetOverlapFraction: 0.30, // 0.05..0.9 (native default 0.30)
    onDeviceDedup: true, // de-duplicate detections across bays on device
    generatePreview: true, // build a stitched display-only preview
    previewMaxDimension: 2048 // max long-edge px of the stitched preview
  }
});
```

Out-of-range values reject with 'INVALID_ARGUMENT', as does 'minBays > maxBays'. An optional 'guidanceStyle: 'classic' | 'glance' (case-insensitive) selects the guidance overlay style.

Attach free-text submit notes to the session before its payload uploads:

```
await Neurolabs.setMultiBaySubmitNotes('Aisle 4, promo end-cap', sessionId);
```

When a multi-bay session completes, the 'cameraClosed' payload carries an extra 'multiBay' summary (forwarded untouched from the native SDK). The per-photo 'captureQueued' / upload events are unchanged:

```
Neurolabs.addListener('cameraClosed', ({ sessionId, cancelled, captureCount, message, multiBay }) => {
  if (!multiBay) return; // single-bay session
  const {
    countsByLabel, // { [label]: count } de-duplicated across bays
    baysCount, // number of bays captured
    dedupTrusted, // false if the cross-bay alignment chain broke
    previewImageFileUri // 'file://...' when generatePreview: true (optional)
  } = multiBay;
  console.log('multi-bay summary', baysCount, dedupTrusted, countsByLabel);
});
```

'dedupTrusted: false' means the on-device de-duplication could not be fully trusted (the cross-bay alignment chain broke, the dedup timed out, or 'onDeviceDedup' was disabled) - counts are then per-bay sums (an upper bound). Treat 'countsByLabel' as an over-count in that case: the backend serves the raw per-image results unchanged (no server-side re-deduplication exists today - the uploaded alignment homographies make one possible later, but on-device dedup is currently the only cross-bay dedup).

There is no partner-facing in-camera bay-mode switch: the Single Multi-bay toggle shown in Neurolabs demo builds is internal debug chrome (gated behind 'NLSDKDebug.internalCameraToolsEnabled' on iOS and its Android equivalent, never enabled in partner builds). The capture mode is controlled purely by config: pass 'multiBay' for a multi-bay session, omit it for single-bay.

7. Post-Processing + Lifecycle Events

```
// Fires when the camera screen is dismissed (Done or Close pressed).
// captureCount is the number of photos queued for upload - NOT the photo data itself.
// To retrieve photo data, use getPhoto() with the captureId from captureQueued (see section 8).
Neurolabs.addListener('cameraClosed', ({ sessionId, cancelled, captureCount, message }) => {
  console.log('cameraClosed', sessionId, cancelled, captureCount, message);
});

// Fires once per photo taken, immediately after each capture is added to the upload queue.
// captureQueued events are buffered while the camera is open and always delivered after cameraClosed.
// Use the captureId here to retrieve the local photo via getPhoto().
// imageFileUri ('file:///...') is present ONLY when file-URI streaming is enabled
// via init({ returnCaptureFileUri: true }) - see 8.1.
Neurolabs.addListener('captureQueued', ({ captureId, queueItemId, sessionId, imageFileUri })
=> {
  console.log('queued', captureId, queueItemId, imageFileUri);
});

// Real-time capture guidance feedback (buffered + coalesced like captureQueued).
// Payload: { rule, tier, timestamp, byDegrees? } where rule is one of
// DISTANCE_TOO_CLOSE | DISTANCE_TOO_FAR | ANGLE_OFF | PANNING_TOO_FAST.
Neurolabs.addListener('guidanceStateChanged', (payload) => console.log('guidance', payload));

// Fires after the Neurolabs server has processed an uploaded photo and returned an analysis result.
// This is a server-side callback - it does NOT fire when the photo is taken locally.
// Payload: { captureId, sessionId, success, message, detectionCount, capturedRegionCount }
Neurolabs.addListener('captureResult', (payload) => {
  console.log('captureResult', payload);
});

Neurolabs.addListener('uploadSucceeded', (item) => console.log('uploaded', item));
Neurolabs.addListener('uploadFailed', (payload) => console.warn('uploadFailed', payload));
Neurolabs.addListener('queueStatusChanged', (status) => console.log('queueStatusChanged', status));
```

8. Retrieving Photos Locally

Photos are stored in the device's app cache after capture. Use 'getPhoto()' to access them by 'captureId' (from 'captureQueued') or 'queueItemId'.

```
const capturedIds = [];

Neurolabs.addListener('captureQueued', ({ captureId }) => {
  capturedIds.push(captureId);
});

Neurolabs.addListener('cameraClosed', async ({ cancelled }) => {
  if (cancelled) return;
  for (const captureId of capturedIds) {
    // format: 'fileUri' returns { uri: 'file://...' } - a temp path in the app cache
    // format: 'base64' returns { base64: '...' } - requires allowBase64PhotoExport: true
    in init()
    const photo = await Neurolabs.getPhoto({ captureId, format: 'fileUri' });
    console.log('photo uri', photo.uri);
  }
  capturedIds.length = 0;
});
```

'deletePhoto()' accepts the same query shape ('captureId', 'queueItemId', or 'responseId') and removes the local file from the cache.

> In file-URI streaming mode (8.1) 'getPhoto({ format: 'base64' })' rejects with > 'UNSUPPORTED_FORMAT' - base64 export is forced off. Use 'format: 'fileUri'', or read the > 'imageFileUri' handed to you on 'captureQueued' directly.

8.1 Capture File-URI Streaming (v1.6.5, OOM fix)

Large multi-capture sessions (wide shelves, dozens of full-resolution JPEGs) could previously push peak memory high enough to get the app OOM-killed, because every captured image was held resident until the host consumed it.

Enable **file-URI streaming** to avoid this - an app-wide, opt-in toggle set once at 'init()':

```
await Neurolabs.init({ apiKey: '<API_KEY>', operationsApiKey: '<OPS_KEY>', returnCaptureFileUris: true });
```

Default is 'false' (unchanged behavior). When 'true':

- Each capture is written to a **stable, backup-excluded on-disk store** and handed to the host as 'imageFileUri' on the 'captureQueued' event, instead of being held in memory. Peak heap no longer grows with the number of captures.
- **base64 export is forced off**: 'getPhoto({ format: 'base64' })' rejects with 'UNSUPPORTED_FORMAT'. Read the 'imageFileUri' instead.
- The retained files **outlive the capture session** - they are kept until the host explicitly releases them, so you can stream/upload them after 'cameraClosed'.

The flag is honored on **both platforms**: Android bakes it into the init-level native configuration; iOS applies it to every capture session.

Releasing retained files

Because the files persist past capture-finish, the host **must** release them once it has finished streaming/uploading each 'imageFileUri', otherwise they linger until a 72-hour TTL sweep reclaims them. Two APIs:


```
const queued = [];
Neurolabs.addListener('captureQueued', ({ captureId, sessionId, imageFileUri }) => {
  queued.push({ captureId, imageFileUri });
});

Neurolabs.addListener('cameraClosed', async ({ sessionId, cancelled }) => {
  if (cancelled) return;
  for (const { captureId, imageFileUri } of queued) {
    await uploadFromFileUri(imageFileUri); // your host-side upload
    await Neurolabs.releaseCapture(captureId); // per-capture release
  }
  queued.length = 0;

  // ...or release the whole session at once instead of per capture:
  // await Neurolabs.acknowledgeCaptures(sessionId);
});
```

- 'acknowledgeCaptures(sessionId)' - bulk release: deletes every retained file for the session. Missing/blank 'sessionId' rejects with 'INVALID_ARGUMENT'.
- 'releaseCapture(captureId)' - single-capture release, for hosts that upload incrementally. Missing/blank 'captureId' rejects with 'INVALID_ARGUMENT'.

Both are no-ops for sessions/captures that kept no file (legacy byte-return mode, or already released), and neither touches the SDK's own upload queue (a separate disk-backed store that streams its uploads independently).

9. Notes

- 'openCamera' is the custom native camera entrypoint.
- Use the strict shelf payload above for full shelf guidance checks and auto-close after a validated save.
- 'liveQualityChecksEnabled=true' is required for pill/rotation/warning/error guidance behavior.
- Keep 'type: 'shelf'', 'guidanceMode: 'strict'', 'showDetections: false', and 'showCapturedRegions: false' for the custom-camera guidance UI path.
- 'guidanceMode: 'guidance'' should only be used as a temporary fallback if a partner explicitly wants looser Android behavior than the parity recommendation.
- 'autoCloseAfterCapture=true' closes after Save from preview/review flow.
- 'Done' appears only in manual-finish mode: 'autoCloseAfterCapture=false' + 'allowManualFinish=true'.
- Use 'minCapturesBeforeDone' to require a minimum number of captures before 'Done' becomes active.
- If 'allowManualFinish=false', 'maxCaptures' acts as a hard limit and capture disables at the limit.
- 'captureQueued' events are buffered while the camera is open and always delivered after 'cameraClosed' - never during an active session.
- Keep base64 transport disabled unless explicitly needed.

Troubleshooting: "Unresolved reference" Kotlin errors on Android build

Symptoms like 'Unresolved reference 'setOperationsBaseUrl'', 'setMultiBaySubmitNotes', 'resourceName', or 'labelsResourceName' during the Android build mean the plugin source is being compiled against a **stale** 'neurolabs-android-sdk.aar' - an old AAR left in 'platforms/android/app/libs/' (which takes precedence) from a previous plugin version.

Fix:

1. Delete the stale AAR and its metadata:
'platforms/android/app/libs/neurolabs-android-sdk.aar' (+ '.meta.json')
2. Re-run 'cordova prepare android' so the plugin re-fetches the AAR matching the plugin's pinned native version.
3. Rebuild.

Since v1.6.3 the Android Gradle script fails the build early with an explicit version-mismatch message (instead of the confusing Kotlin errors) whenever the resolved AAR's version does not match the plugin's 'neurolabs.sdkVersions.android'.

Upgrading from a pre-1.6 plugin - queue durability

On pre-1.6 native SDKs the upload queue was wiped by the very bridge message that carries your API config. Sending 'NL_Config' / the 'api_config' action (which you do to set credentials + per-visit 'accountId'/'storeId'/'visitId') triggered a full queue clear on the native side - metadata **and** the stored JPEGs - so any not-yet-uploaded captures from a previous visit were deleted at the start of the next one, regardless of network.

- **Already-lost captures cannot be recovered.** They were removed from device storage by the reconfigure-wipe - even sessions that were never lost to timeouts were destroyed this way. There is nothing to migrate.
- **Upgrading stops the loss going forward.** From native v1.6.1+ (plugin bundling native 1.6.1) the config message no longer clears the queue; captures persist across reconfigures, backgrounding, and restarts. Only an explicit 'clearUploadQueue()' removes them.

Upgrade to a plugin bundling native **1.6.1** (1.6.3+ recommended) and set 'operationsApiKey' at init so the preserved queue can deliver.

On-Device Recognition + Tap-to-Product-Card (v1.7.1, Android)

Config-gated per account: your operations API key resolves your org's recognition config and vector dataset - enabling it is a backend flip, no app change. When active, 'captureResult' events carry 'recognitionMatches: { <detectionId>: { catalogItemId, similarity } }' (each detection object carries the correlating 'id'); resolve them for your card UI:

```
const details = await Neurolabs.getProductDetails(  
  Object.values(payload.recognitionMatches).map((m) => m.catalogItemId)  
);  
// details["<uuid>"] { canonicalUuid, name?, brand?, flavour?,  
//                      containerSize?, barcode?, thumbnailUrl? }
```

Absent keys are unknown products - render the raw UUID + similarity as the fallback. Results are cached natively for 24 h. iOS parity for the recognition provider arrives in a follow-up release; 'getProductDetails' works on both platforms today.